PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

Castillano's Backyard
Integrated Livestock Ordering System

A Capstone Project

Presented to the Faculty of the

College of Information Technology Education

PHINMA University of Iloilo

In partial fulfillment

of the Requirements for the degree

Bachelor of Science in Information Technology

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Approval Sheet

In partial fulfillment of the requirements for the Bachelor of Science in Information Technology, this Capstone Project entitled **"Castillano's Backyard Integrated Livestock Ordering System"** prepared and submitted by Phoebe Mae Almario, Mary Mae Babac, Kenth Axell Castillano and Mae Christil Joy Sumbing who are hereby recommended for corresponding final evaluation.

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Acknowledgement

We would like to express our heartfelt thanks and sincere gratitude to all the individuals who contributed to the success and completion of this project. Their support, guidance, and encouragement played a significant role in making this project possible.

First, we are very thankful to God for giving us the strength, guidance, and determination to complete this Capstone Project.

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To the Castillano's Backyard especially to **Mr. Martin Castillano Jr.** and his family for allowing us to create this project for their business may the mighty God Bless You thank you for your all out support since day 1 that we planned to create this system until the day we almost finish the project and also MRS. Gloria Gicole, the representative of Castillano's backyard, who joined us during our defense day and also supported us. Thank you very much ma'am!

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Abstract

Modern technology is overpowering these days. People rely on online shopping, like buying things they want in just one click and having them delivered straight to their house. Everything is made convenient for the customer; easy to access, order and purchase. This has made the customer's life easier.

Castillano's Backyard's problem was that they use manual management of their inventory and sale of their product, which include livestock like live pigs and chickens and the eggs and artificial insemination for pigs.

The integrated livestock ordering system in Castillano is a livestock ordering website that helps the Castillano's track their sales and profit from time to time by using charts and have a monthly sales record that shows the records of the order.

The developers proposed this system will be a great help for the client and that this website will also help the Castellano to be known as the livestock seller and artificial insemination services.

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COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

TABLE OF CONTENTS

LIST OF FIGURE

LIST OF TABLES

PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

Chapter 1

Introduction of the Study

The project aims to design and develop an e-commerce website for Castillano's Backyard, which will facilitate inventory management with record keeping and the generation of sales reports with the application of data visualization. This system aims to improve inventory management and record tracking, ensure the accuracy of records, and apply data analytics to improve the business' efficiency, productivity, and performance.

This project will focus on developing an e-commerce website with user-friendly interface for Castillano's Backyard that will allow the customers to order livestock and products online. It will allow the owner to manage their inventory efficiently and track and monitor their sales records. It will use a database management system to store product and sales data. The system will incorporate data analytics to provide reports on sales, providing the business with important data to make informed decisions. This system will be designed for the business to improve its management system to its efficiency, productivity, and growth.

By the end of this project, the system will help Castillano's Backyard improve its inventory management and record keeping to improve sustainability with efficiency, and profitability, to reduce waste and production costs.

Technical Background of the Study

E-commerce refers to the buying and selling of goods and services over the internet. It relies on a combination of digital technologies, communication networks, and information systems to facilitate online transactions between businesses and consumers.

An inventory management system is made to track, monitor, and manage inventory levels. Businesses often use inventory management systems to track and control stocks before they are sold out. The system aims to help and maintain enough stocks to their customers' needs and lessen the inventory carrying cost.

Data analytics is a technology that uses math, statistics, and machine learning to find patterns in data. It is created to discover, interpret, and to share insight. Businesses use data analytics to turn data into insight for making informed business decisions and strategic planning.

Implementing an inventory management system with analytics is vital for businesses to manage and track their inventory and use analytics to gather insights to make informed decisions.

Key technologies involved in such systems include:

1. **Database Management Systems (DBMS)**- Systems like MySQL store, organize, and manage inventory data efficiently.
2. **Data Visualization Tools** - Visualization tools generate charts and reports, allowing for easier data interpretation.
3. **Data Analytics**- It aims to collect data and to emphasize analysis for more insights that can help farm owners/managers make informed decisions, optimize resource use, improve efficiency, and boost the health and productivity of their livestock.

Statement of the Problem

Poultry and swine industries often rely on manual management systems, which can lead to inefficiency, inaccuracy, discrepancies, and waste. This project aims to address the following problems:

1. **Inefficient Inventory Management**
   Many poultry and swine farms depend on manual methods to track livestock, products, and sales. This results in inaccurate records, stock shortages, overstocking, and unnecessary waste of resources.
2. **Data Collection and Reporting Inaccuracies**
   Manual inventory tracking produces unreliable data on livestock and sales. The lack of real-time visibility makes it difficult to make informed decisions and properly allocate resources, increasing the risk of financial loss and operational inefficiency.
3. **Lack of Real-Time Data Monitoring**
   Manual data collection prevents continuous monitoring of backyard operations. This leads to delayed responses to critical issues such as disease outbreaks or feed shortages, negatively affecting farm productivity and animal welfare.

Objectives of the Study

This project aims to design and develop **Castillano's Backyard: Integrated Livestock Ordering System** that will:

1. **Develop an e-commerce platform** that enables customers to browse livestock products and place orders online efficiently.
2. **Implement an inventory management system** to monitor, update, and manage livestock resources and product availability in real time.
3. **Incorporate data analytics and reporting features** to generate real-time sales reports and provide insights for effective inventory and business decision-making.
4. **Design a user-friendly interface** that ensures ease of navigation, visual clarity, and an improved user experience to attract and retain customers.

Significance of the Study

The findings of this project will focus on Castillano's Backyard by providing an inventory management system to address the issue of relying on manual processes. The system will visualize the sales, expense,and profit to assist in decision-making and improve efficiency, reducing errors, and business growth.

The key benefits of this study include:

1. **Improved Efficiency:** inventory tracking and management will reduce the time spent on manual tasks, allowing businesses to focus on more critical activities.

2. **Reduced Errors:** The system will minimize human errors by using technologies like automated generation of financial reports to be sure of the accuracy and integrity of data.
3. **Better Decision-Making:** With analytics, businesses can make informed decisions regarding inventory, stocks, and demand forecasting.

Scope and Limitations

The **Castillano's Backyard: Integrated Livestock Ordering System (ILOS)** is a web-based platform designed to make ordering livestock products online easier. The system allows customers to browse available poultry products and place orders using a simple interface. At the same time, it lets the administrator manage orders, update product information, and track transactions effectively. It also reduces errors and improves accuracy compared to older manual processes.

1. **Cart And Order System -** Customers can place orders directly through the website, eliminating the confusion and errors of manual recording. Admins can easily track, approve, and manage all customer orders in one place.
2. **Product Availability -** The system allows customers to browse products, check availability from anywhere — increasing convenience and improving customer satisfaction.
3. **Sales Tracking & Visualization -** The system provides the admin with visual sales reports (monthly), helping the business monitor its sales and identify trends without using manual spreadsheets.

4. **Automated Records -** Manual methods often lead to missing orders, duplicate entries, or miscalculated profits. The system reduces these risks by automating inventory presentation and order intake.

While the system provides essential e-commerce functionality for livestock management, several limitations were identified during development:

- The system currently does not support online payment transactions that are confirmed manually after order approval by the admin.
- The system is accessible through web browsers and does not include a mobile application version.
- Delivery tracking and logistics integration Customers must coordinate pickup or delivery directly with the farm owner.
- The system relies on stable internet connectivity, users in remote areas may experience slow loading or limited access.

Despite these limitations, the system meets its primary goal to streamline livestock ordering, improve record accuracy, and enhance communication between the farmer and customers through a simple and accessible online platform.

Definition of Terms

1. ***Data Analytics*** – The use of analytical tools and techniques to predict demand, optimize inventory levels, and support better decision-making in inventory management.
2. ***Inventory Management*** – The systematic process of managing the ordering, storage, tracking, and usage of a company's products to ensure availability and efficiency.
3. ***Inventory*** – The collection of raw materials, goods, or products held by a business for sale, production, or distribution purposes.
4. ***User Interface*** – The visual and interactive components of the system that allow users to navigate and perform functions, including menus, buttons, and data displays.
5. ***User-friendly Interface*** – A system design focused on ease of use, enabling users to navigate and operate the inventory system efficiently with little to no training.
6. ***Automation*** – The application of technology to carry out tasks with minimal or no human involvement, improving efficiency and reducing manual effort.

Chapter 2

Review of Related Literature

This chapter reviews relevant literature and studies that support the development of the ***Castillano's Backyard: Integrated Livestock Ordering System***. The discussion covers topics related to information systems and e-commerce. These concepts provide the foundation for building a system that improves product management, order processing, and overall business efficiency.

Management information systems (MIS)

Haag(2009) explains that Management information systems (MIS) deals with the planning for,development,management and use of information technology tools to help people perform all tasks related to information processing and management So MIS deals with the coordination and use of three very important organizational resources information people and information technology stated another way,people use information technology to work information .and to do so they are involved in MIS.ideally of course people use technology to support the goals and objectives of the organization as driven by competitive pressure and determined by appropriate business strategies MIS helps them to do this. Collective information about your customers, your competitors, your business partners, your competitive environment and your own internal operations that give you the ability to make effective, important and often strategic business decisions.

Strategies to predict e-commerce inventory and ordering planning.

Rahman and Casonovas(2021) states that the rise of e-commerce has significantly transformed traditional retail operations, making online shopping a dominant force in modern commerce. According to Rahman and Casonovas (2021), e-commerce has revolutionized trade by creating an information-rich platform that enables firms of all sizes to engage in the digital market. The appeal of e-commerce lies in its low initial investment, minimal manpower requirements, quick setup, reduced inventory costs, and the ability to facilitate instant communication and high transaction visibility. These factors have contributed to the rapid expansion and popularity of online marketplaces.

Accurate prediction of merchandise demand is essential for success in the e-commerce sector. Demand forecasting plays a critical role in inventory and order planning, enabling businesses to meet customer expectations effectively. Rahman and Casonovas (2021) emphasize that understanding market trends and consumer behavior is key to developing reliable inventory management strategies.

Furthermore, the use of web-based systems in e-commerce allows for seamless buying and selling of goods and services, as well as the efficient transfer of information, data, and funds. These benefits have enhanced the overall value proposition for both sellers and buyers. The convenience of timely delivery and streamlined inventory processes motivates consumers to engage in frequent online purchases, further driving the growth of the e-commerce industry.

References

Haag, S., & Cummings, M. (2009). *Management information systems for the information age* (7th ed.). McGraw-Hill Higher Education. ISBN 978-0-07-128796-8

Rahman, M. A., & Casonovas, L. (2021). Strategies to predict e-commerce inventory and ordering planning. In M. Khosrow-Pour (Ed.), Handbook of Research on Intelligent Techniques and Modeling Applications in Marketing Analytics. ISBN 1799889580

Chapter 3

Methodology

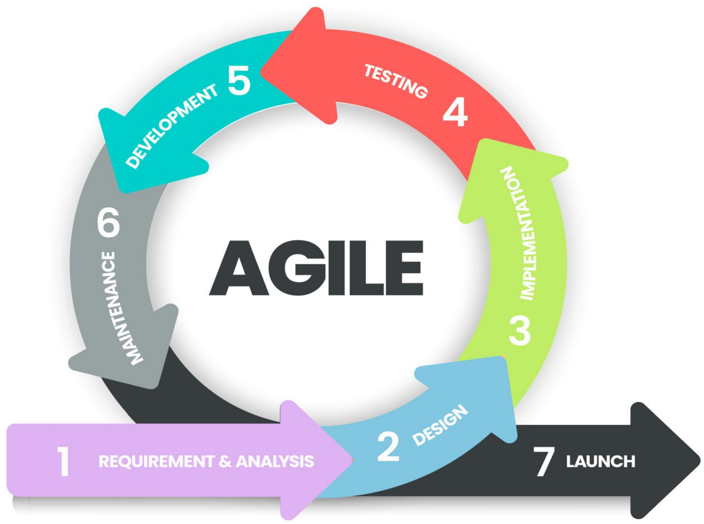
Research and Development Stage

The *Castillano's Backyard: Integrated Online Livestock Ordering* is designed to improve efficiency in ordering, inventory management, and customer interaction for a small-scale backyard livestock business. To ensure the system meets real-world needs, the project uses the **Agile development methodology**, allowing for flexible, user-centered, and iterative development.

This approach promotes adaptive planning, early delivery, and continuous improvement. It involves close collaboration between the development team and the end users—farm owners, employees, and customers—throughout the entire project lifecycle.

Agile Methodology

The development of the *Castillano's Backyard: Integrated Online Livestock Ordering System* is based on the **Agile Software Development Methodology.** Agile was chosen for its ability to deliver working components in short, manageable increments while adapting to stakeholder feedback. It emphasizes collaboration, flexibility, and rapid iteration — ideal for a community-centered e-commerce system that evolves based on actual use.



The Agile process for this system followed the sprint outlined below:

1. Requirement & Analysis

During this phase, the project team collaborated with the Castillano's Backyard to identify the problems encountered in traditional inventory management and order processing. Common issues included:

- Misplaced or inaccurate records
- Delays in order fulfillment
- Inability to monitor real-time sales and profit

The team documented the system requirements, including:

- Online ordering interface
- Admin dashboard with sales and order reports

- Manual order approval system

2. Design

The design phase involved creating the system’s structure, user interface, and user experience plans. This included:

- User-friendly frontend design that allows customers to browse poultry products and place orders easily.
- Backend system using PHP and MySQL for data processing and storage.
- Admin panel design for managing products, monitoring orders, and overseeing system activities.
- Database structure designed to store and manage product details, customer information, messages, and order records.
- System architecture planning to support future improvements and new features.

3. Implementation

In this phase, the development team translated the designs into working code. Each module was implemented according to the sprint plan, which included:

- Customer interface: browse products, place orders, contact admin
- Admin Interface: manage the system through a dashboard. This includes viewing pending orders, approving , canceling orders, and monitoring basic system activity.
- Product Management: allows the admin to add new products, update product information like price and description, and manage product listings directly from the admin panel.

The system was developed using PHP, MySQL, HTML/CSS/JS, and Python Flask (for future analytics integration).

4. Testing

To ensure functionality and reliability, the system underwent several layers of testing:

- **Unit Testing:** Verified each module (e.g., order checkout, login)
- **Integration Testing:** Ensured components worked smoothly together
- **User Acceptance Testing:**the client tested the system and provided feedback

5. Development (Sprints)

The system was built in iterative sprints, allowing key features to be implemented and tested quickly. This sprint-based development allowed feedback from the client to be incorporated regularly. Each sprint delivered working components ready for use or demonstration.

6. Launch / Deployment

After testing all major modules, the system was deployed in a live environment for both customers and the administrator. Deployment activities included:

- Final setup and configuration of the server and database.
- Orientation and walkthrough for the administrator on how to use the system.

- Initial uploading of product data, with the admin able to add, edit, and manage products directly through the system.

7. Maintenance

Following deployment, the system entered the maintenance phase. The development team remains responsible for:

- Fixing reported bugs
- Monitoring performance and server stability
- Preparing for future updates, such as online payment, mobile app version, and machine learning enhancements

Data Collection Techniques

Different kinds of data collection methods were used to obtain the necessary information to develop the Castillano's Backyard: Integrated livestock Ordering System These techniques were designed to address the challenges of selling livestock and managing inventory online while ensuring that the system aligns with the needs of both the owner and customers. The following methods were utilized:

1. Surveys

Surveys were used as the primary tool to gather quantitative data from potential customers and the system administrator. The questionnaires focused on understanding customer behavior in purchasing livestock products such as pigs, chickens, eggs, and artificial insemination services online. It also included questions about payment preferences, user experience expectations, and trust factors in digital livestock transactions.

For the admin, the survey gathered input on stock management issues, profit tracking, and challenges in handling multiple products. This data guided the design of features that prioritize simplicity, security, and convenience.

2. Interviews

In depth interviews were conducted with the owner of Castillano's Backyard and several regular customers to collect qualitative insights. The goal was to understand their day-to-day processes, pain points in traditional livestock selling, and expectations for an online platform.

The discussions provided valuable feedback on how to present livestock listings, manage customer communication, and handle payments securely. These insights directly influenced the development of an intuitive admin dashboard and customer-friendly interface.

3. Observations

Direct observations were carried out at Castillano's Backyard to study how inventory and sales are managed manually. This included observing how livestock and egg stocks were tracked, how orders were fulfilled, and how records were maintained.

By observing these processes, the development team identified areas prone to human error, such as delayed updates and inaccurate stock counts, and proposed automation features within the system to improve accuracy and efficiency.

4. Document Analysis

Existing business records, transaction logs, and product inventory lists from Castillano’s Backyard were analyzed to better understand current operations. Additionally, references to other e-commerce and livestock management platforms were reviewed to identify best practices in online product management, payment handling, and customer service.

This analysis helped refine the database structure and ensured that the system meets the real operational needs of a small-scale poultry and livestock business.

Justification for Selecting the Agile Methodology

Agile technique was chosen for an online livestock ordering and inventory management system project due to its flexibility, ongoing customer input, and continuous development.

ASPECTS	AGILE METHOD	TRADITIONAL APPROACH
Development Style	Short sprints of iterative and incremental development were used to provide functional modules (such ordering, inventory, and admin administration) ahead of schedule.	Linear or phased, where the complete system is delivered only at the end.

Flexibility	Very flexible when it comes to adding new livestock varieties or changing price features to meet the needs of new clients.	Limited adaptability; modifications following requirement collection are expensive and time-consuming.
User Involvement	Active collaboration with the farm owner and potential customers to test, validate, and refine the system.	Limited participation; users only provide input at the start and end of development.
Delivery	Every stage, there should be regular updates with useful features that can be tested.	input by the user is delayed by a single final delivery after all steps.
Risk Management	Risks are detected and resolved early during each sprint, reducing project delays.	Risks are often identified late, requiring more time and resources to fix.

Testing	Continuous testing ensures reliability and accuracy	Before moving forward, continuous testing makes sure that new features like the admin dashboard, order system, and messaging work as intended.
Collaboration	supports close collaboration between the client and developers to ensure accurate feature connection and business logic	Moderate – structured roles and limited communication
Adaptability to Change	Excellent – quickly incorporates feedback and new needs	Poor – changes require revisiting completed phases
Efficiency	Reduces errors, rework, and speeds up development	Can be time-consuming and costly if errors arise late

System Architecture and Design

Overview

The Castillano's Backyard: Integrated Online Livestock Ordering System is a web-based system designed to automate the management of livestock orders, inventory tracking, and customer transactions for a small-scale backyard business. The system aims to streamline operations by providing a centralized platform for ordering, sales monitoring, and communication between admin and customers. The system is modular and scalable, consisting of three main layers: presentation, application, and data. It is web-based, accessible through desktop and mobile browsers, and developed using PHP, HTML, CSS, JavaScript, and MySQL. The system supports future expansion, such as cashless payment integration and mobile responsiveness. The system ensures efficient and accurate processing of orders, sales, and livestock records, meeting operational needs and improving customer experience.

System Architecture

The **Castillano Backyard: Integrated Livestock Ordering System (ILOS)** is designed using a **two-tier architecture**, which separates the **Presentation Layer** and the **Data Layer.** This design approach ensures that the system is maintained, while also providing a responsive and user-friendly experience for both customers and the admin.

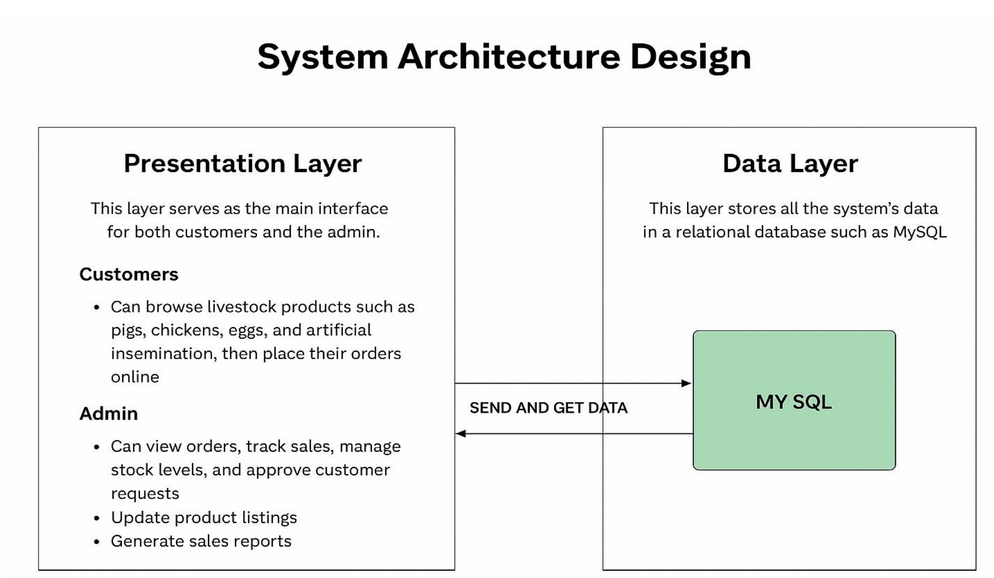
1. Presentation Layer

allows communication between users and the system and acts as the main user interface for all users.guaranteeing a response . user-friendly design that is usable with all current web browsers. The customers can view available livestock products such as pigs, chickens, eggs, and artificial insemination services, also they can Place orders for available products and view their order history. The admin side can Add, update, and delete livestock product listings. managing the sales level and monitoring the stock availability

Data Layer

The Data Layer securely stores, manages, and retrieves system information using MySQL as the database management system. It contains structured tables for products, customers, orders, and sales. The Data Layer ensures data integrity through relational constraints, security through restricted access and authentication protocols, and regular database backups to prevent data loss. This ensures efficient data handling and quick response times during operations.

Figure 2. System Architecture Diagram



System Components and Modules

1. Dashboard Module- Displays a summary of the system’s key information and performance metrics.

Key Features:

- View total number of products and customer orders
- Display overall sales summary based on transactions
- Show recent orders and system activity updates
- Provide quick access to key modules such as products, orders, and customer management

2. Sales and Transaction Module - processes customer orders and manages payments.

Key Features:

- Add and manage customer orders
- Record order details and total amounts
- Maintain transaction records for monitoring and review
- Update order status and reflect changes in product listings

3. Inventory Management Module- Keeps track of available livestock, eggs, and other stock.

Key Features:

- View all product listings and their available quantities
- Update product stock information as needed
- Reflect stock changes based on customer orders
- Maintain records of product availability for monitoring purposes

4. Profit and Reporting Module- Generates financial reports and calculates total sales and profit.

Key Features:

- Calculate total sales based on recorded orders
- Generate simple reports from available transaction data
- Display summarized information for monitoring business performance
- Provide an overview of sales activity for the administrator

5. User Management Module- Controls system access and user accounts.

Key Features:

- Admin login authentication for secure system access
- Password security and account protection
- Controlled user access based on roles
- Ability to manage and update account information

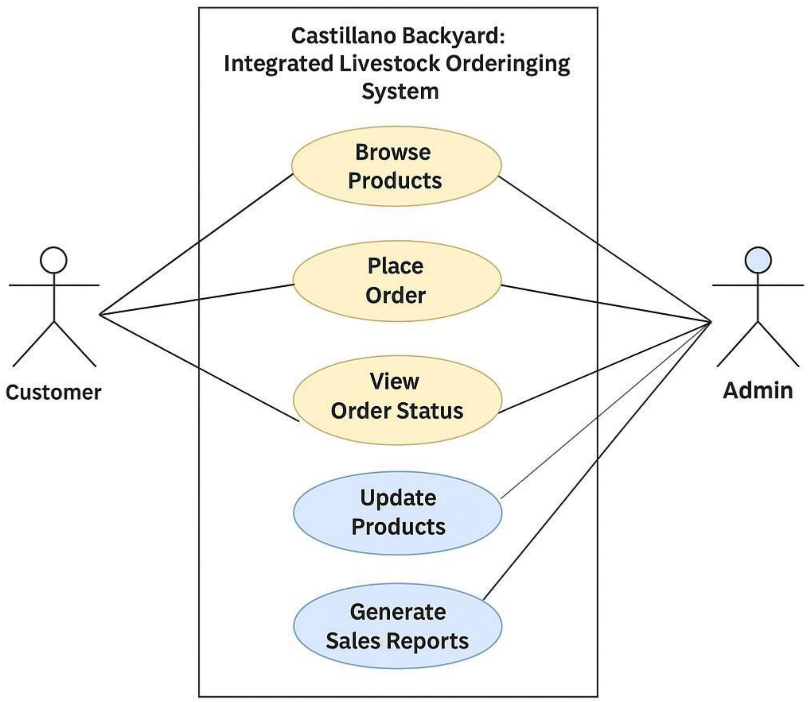
6. Customer Management Module- Maintains customer records and transaction history.

Key Features:

- Register and update customer information
- View transaction history for each customer

Figure 3. Use Case Diagram

Use Case Diagram



Data Flow Diagram

Data Flow Diagram Level 0

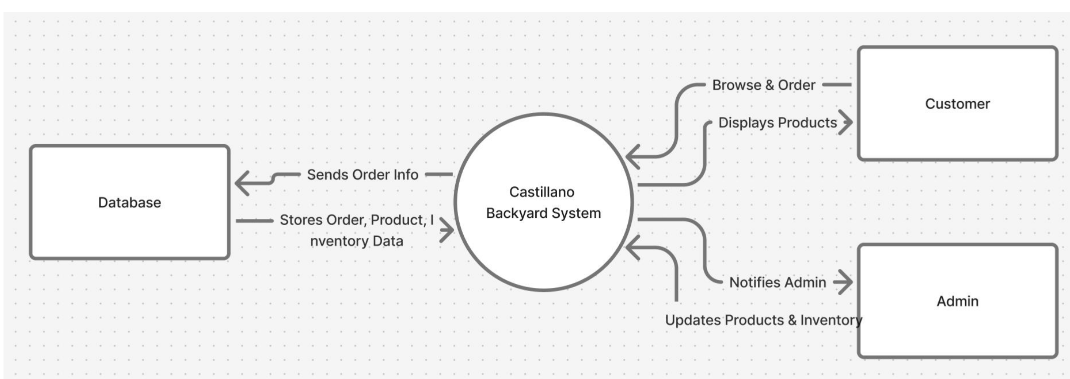
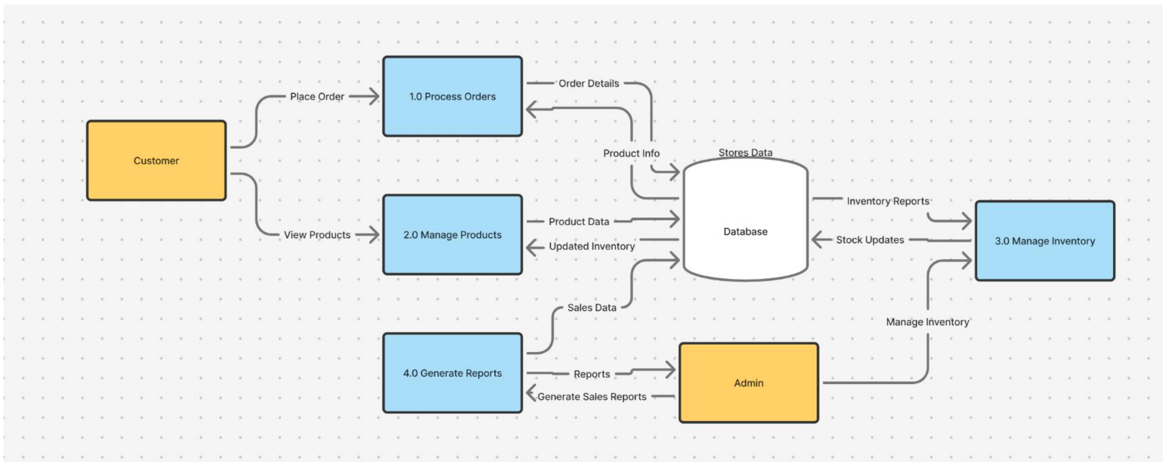


Figure4: Data Flow Diagram Level 1



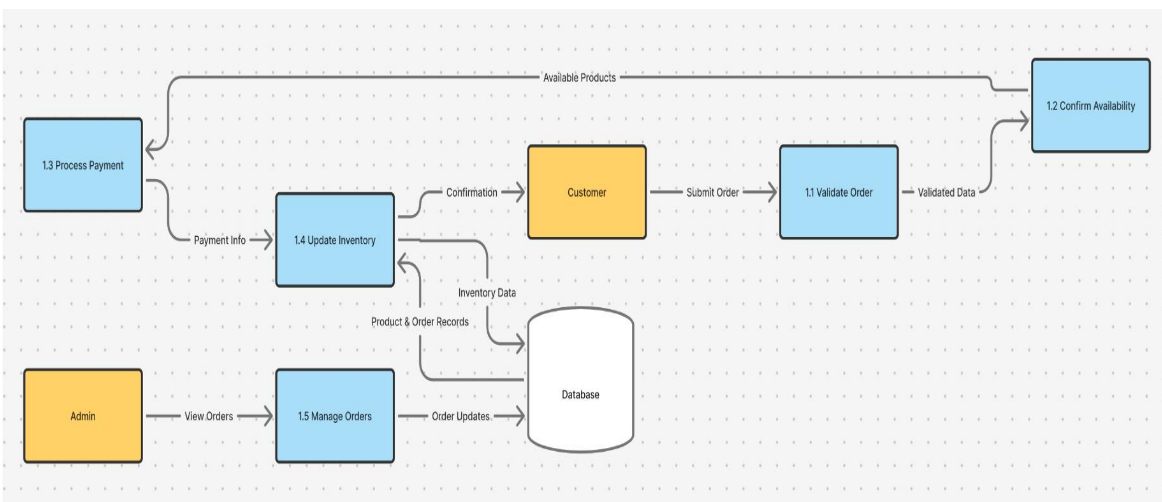


Figure 5: Data Flow Diagram Level 2

The DFD Level 2 further elaborates on the “Process Orders” function presented in the Level 1 diagram. It illustrates the detailed flow of data between the system’s internal processes, data stores, and external entities involved in handling customer orders. This level of detail ensures that each step of the order processing workflow is clearly defined and traceable.

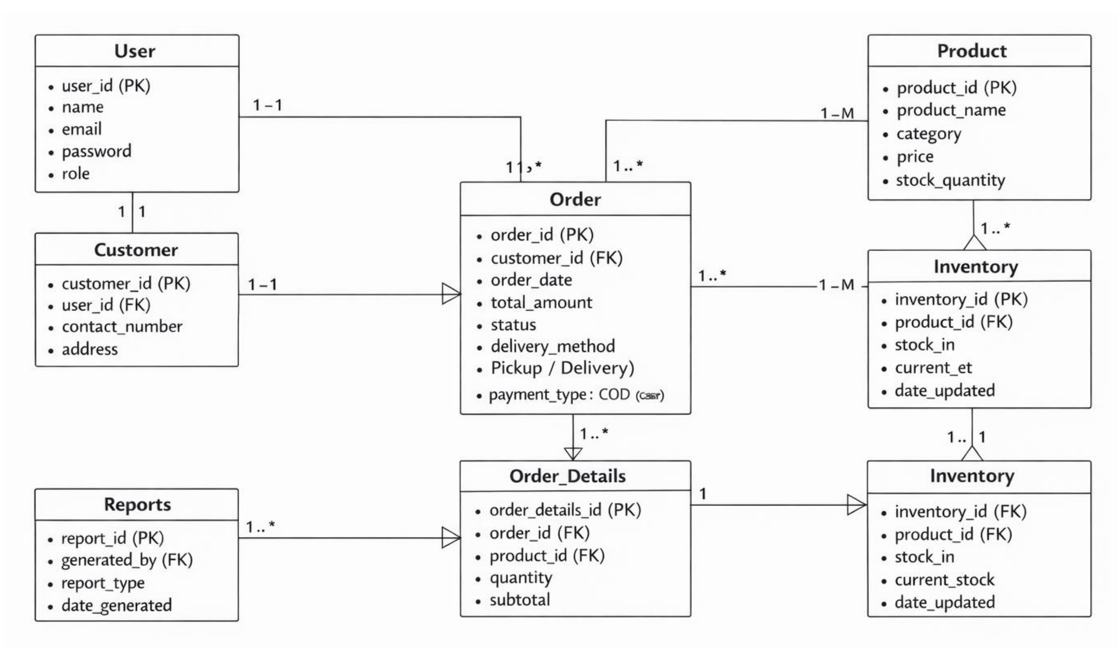
When a customer places an order through the system, the process begins with the “Validate Order” activity, where the system checks product availability and verifies order details such as quantity, pricing, and customer information. Once validated, the data flows to the “Update Inventory” process, which automatically deducts the ordered quantity from the inventory database to maintain real-time stock levels.

After inventory updates, the order details are sent to the “Record Transaction” process, where all relevant information — including customer name, product type, quantity, total cost, and payment method — is stored in the Sales Database. This ensures that both sales and inventory records remain synchronized and traceable for reporting and analytics purposes.

Finally, the system generates an “Order Confirmation” message that is sent back to the customer. This confirmation contains order details and serves as proof that the transaction was successfully recorded. Simultaneously, the admin can access updated reports and transaction logs through the dashboard for monitoring sales and performance.

Overall, the Level 2 DFD emphasizes the automation of order handling, accuracy of real-time stock updates, and the seamless communication between the customer, admin, and system databases — leading to a more efficient and reliable livestock ordering process.

Entity Relationship Diagram



Data Base Design

The system's database uses MySQL and has a relational structure to manage data efficiently. It includes tables for users, products, cart, orders, order details, payments, and messages. Each table has specific fields with unique identifiers like primary keys and connects through foreign key relationships to keep data intact and reduce redundancy. This setup allows for precise tracking of inventory, customer orders, transactions, and system interactions, ensuring smooth and reliable performance.

User Interface Design

The User Interface (UI) of the *Integrated Livestock Ordering System* was designed to be simple, user-friendly and accessible. It helps administrators, employees, and management to easily access and manage operations, including data handling and sales tracking. The clean layout and simple navigation let users finish tasks quickly and effectively.

The system has a secure login page that requires a username and password. option and role-based access. Users can view their order history and check order statuses, including pending, approved, shipped, or delivered.

The dashboard acts as the main interface, giving an overview of important information such as available products, stock levels, order summaries, and sales data. It has a navigation menu or sidebar that lets users access different sections of the system, such as product management, cart, orders, reports.

Security Consideration

Security is crucial in the ILOS because it deals with sensitive business information like financial records, production reports, and customer data. The system has various security features to protect the confidentiality, integrity, and availability of this information.

Users must log in with a valid username and password, and each account is linked to a specific role, like Administrator or Staff, determining their access rights. This means only authorized users can view or change certain data. Passwords are stored in an encrypted format to enhance security, and the system checks login details to block unauthorized access. It also logs users out after a period of inactivity to prevent unauthorized use.

Input validation helps protect against security risks like SQL injection and cross-site scripting. Access to sensitive areas like financial reports is limited to admin users. The system tracks user activities to allow for auditing and accountability. Regular backups of the database are made and stored securely to prevent data loss. Encrypted communication protocols are used when the system is online, ensuring that all information remains safe from unauthorized access or corruption.

Technology Stack

The ILOS was developed using modern tools and technologies to ensure efficiency, scalability, and user-friendliness across all modules.’

1. Front-End Technologies

The **front-end** or **client-side** of the system is responsible for the visual presentation and user interaction. It uses HTML for web page structure, CSS for styling and layout design, and JavaScript for interactivity and dynamic features. The Bootstrap Framework helps design mobile-friendly layouts with ready-made styles and components.

2. Back-End Technologies

The **back-end** or **server-side** The back-end of a system plays a crucial role in managing the overall logic, processing data, and facilitating communication with the database. It primarily utilizes PHP as the main server-side scripting language, which is essential for handling incoming requests, processing data efficiently, and managing user authentication securely. This architecture ensures that all interactions with the database are conducted smoothly and that the application remains responsive to user inputs.

3. Database Technologies

MySQL is the main database management system, which holds different types of information, including product details, customer information, and order records data for reliability and security. phpMyAdmin assists in managing MySQL through table creation and data backup facilities. This setup ensures good data organization and accessibility, which is important for the system's overall functionality.

4. Development Tools

- **Visual Studio Code / Sublime Text:** Code editors used for writing, debugging, and maintaining system source code.

- **XAMPP:** A local server environment that includes Apache, MySQL, and PHP, allowing for local development and testing.

5. Server and Hosting

- **Apache Web Server:** Handles HTTP requests and responses, allowing users to access the web-based application.
- **Localhost / Web Hosting (e.g., InfinityFree, 000Webhost):** Used for deploying the system online and testing accessibility across different devices.

Data Gather Techniques

Survey Questionnaire For System Assessment

The purpose of this survey was to obtain user feedback on "Castellano's Backyard." The information gathered was used to determine how the system will be improved based on the user feedback.

Demographic Information

Name: ____________________
Age: _______

Age:
☐ 18-25 ☐ 26-35 ☐ 36-45 ☐ 46 and above

Gender:
☐ Male ☐ Female ☐ Prefer not to say
Frequency of Using the System:
☐ Daily ☐ Weekly ☐ Occasionally ☐ Rarely

Please rate your level of agreement with the following statements based on your experience using the system.

Scale:

5 – Strongly Agree 4 – Agree 3 – Neutral 2 – Disagree 1 – Strongly Disagree

1. The system is easy to use and navigate.
2. The interface is clear and visually organized.
3. I can easily find the features or functions I need.
4. The system accurately records poultry data such as feeds, stocks, and production.
5. The reports generated by the system are useful and complete.
6. The system's features meet the needs of poultry management.
7. The system performs tasks quickly without delay or error.
8. Using the system reduces manual work and saves time.
9. The system keeps my information and account secure from unauthorized access.
10. Overall, I am satisfied with the ILOS and would recommend it to others.
11. What feature of the system do you find most useful?
    ____
12. What problems or challenges have you encountered while using the system?
    ____
13. What suggestions do you have for improving the system?
    ____

Sampling Methods for Respondents

The researchers used a **purposive sampling method** in selecting the respondents of the study. This method was chosen to ensure that only individuals who have direct experience using the integrated livestock ordering system were included. The respondents consisted of poultry owners, staff, and administrators who regularly used the system in their operations. This approach allowed the researchers to gather accurate and relevant feedback from knowledgeable users. Through this sampling method, the study obtained reliable data to evaluate the system's usability, functionality, and overall effectiveness.

Data Analysis Techniques

The information gathered from the ILOS users were quantitatively analyzed to identify the level of effectiveness, efficiency, and user satisfaction of the system. The researchers used a Likert Scale questionnaire for assessing the respondents' perceptions of the usability, accuracy, performance, security, and overall functionality of the system. Each answer was given a numeric value and calculated with the weighted mean formula to derive the average ratings of each indicator. The findings were then translated with descriptive scales to determine the strength and weaknesses of the system. With these data analysis methods, the researchers were able to give an explicit and unbiased evaluation of the overall quality and user acceptance of the ILOS

Ethical Conversation

The Integrated livestock ordering system was created to follow strict ethical standards in technology use. It keeps all user data, like poultry records and account information, private and secure. Only authorized users can access specific parts of the system based on their roles, which helps prevent misuse. The system does not collect unnecessary personal information and only uses data for proper business management. It supports responsible technology use for fair and transparent poultry operations. Additionally, it uses encryption and role-based access control to protect data from unauthorized access. Overall, the system ensures that all information is kept private and secure.

Chapter 4

Results and Interpretation

Summary of the Data Sources and Analysis Methods Used

The data in this chapter came from user surveys, system testing, and observations made during the evaluation phase of the Integrated Livestock Ordering System (ILOS). The researchers used both quantitative and qualitative methods to assess how well the system performed.

Quantitative data were gathered through Likert-scale questionnaires completed by respondents who are Castillano's customers. The researchers analyzed this data using frequency counts, percentage distribution, and weighted mean to find the level of agreement for each indicator.

Qualitative data were collected through feedback and observations to back up the numerical findings. The analysis looked at usability, functionality, efficiency, accuracy, security, and overall user satisfaction.

Task Manager

Week	Task	Programmer	UI/UX Designer	Documenter
1.	Project Planning & Requirements gathering	✓	✓	✓
2.	Research & System Design Planning	✓	✓	✓
3.	Use Case & DFD Creation	✓	✓	

4.Database Design & schema creation	✓	✓	
5.ui Wireframes &mockups	✓	✓	
6.Frontend Development begins	✓	✓	✓
7.Backend Development begins	✓	✓	✓
8.Integration of frontend and backend	✓	✓	✓
9.System testing & debugging	✓	✓	✓
10.user testing & feedback collection	✓		
11.final revisions (code + ui + content)	✓	✓	
12.documentation finalization & formatting	✓	✓	✓

Presentation of results

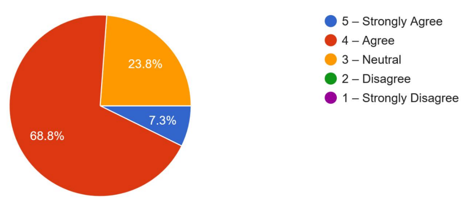
The Integrated Livestock Ordering System (ILOS) led to clear improvements in managing backyard operations. Users reported that the system is efficient, easy to use, and dependable. The system cut down on manual work, increased the accuracy of data recording, and improved overall operational efficiency. Respondents mostly shared positive feedback about its usability, functionality, and security.

Data Tables, Graphs and figures

Metric	Before System	After System
Data Recording Accuracy	Manual/ Prone to error	Accurate
Task Completion Time	Slow	Fast
User Satisfaction	Moderate	High
Human Error Rate	Moderate	Minimal

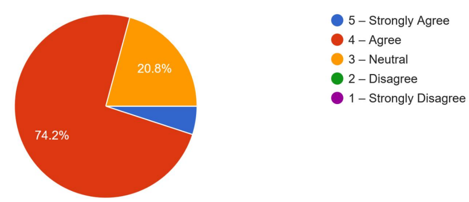
The system accurately records poultry data such as stocks, and production.

260 responses



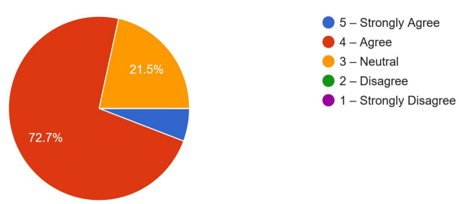
The reports generated by the system are useful and complete.

260 responses



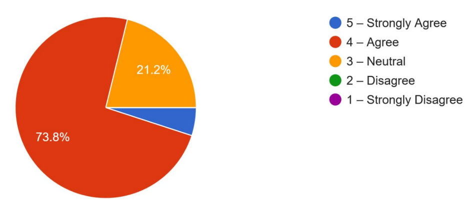
Using the system reduces manual work and saves time.

260 responses



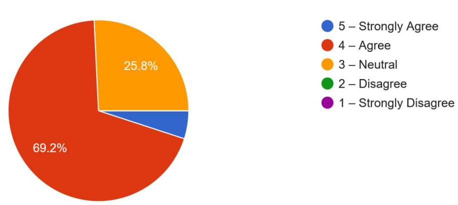
The system keeps my information and account secure from unauthorized access.

260 responses



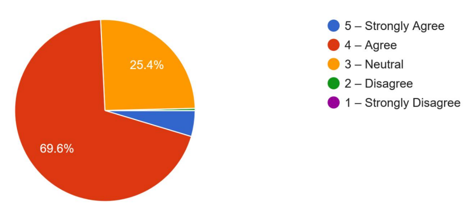
The system's features meet the needs of poultry management.

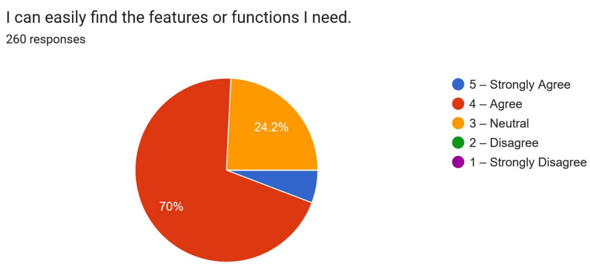
260 responses

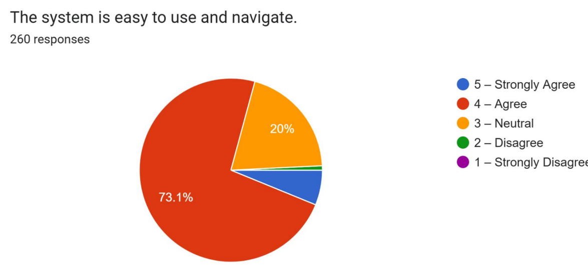


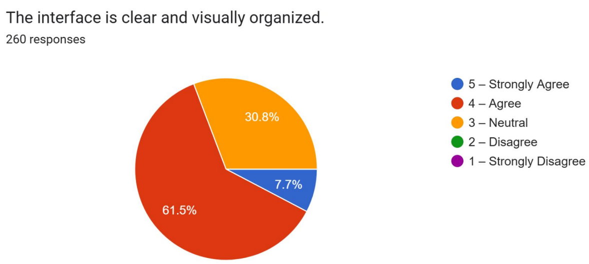
The system performs tasks quickly without delay or error.

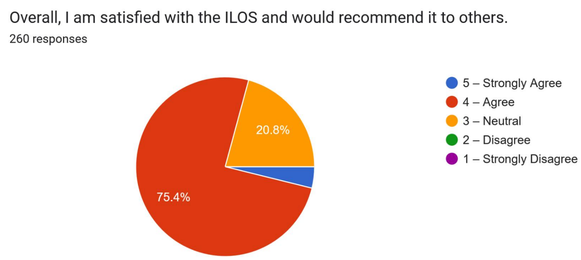
260 responses











The graphs shows the users mostly agrees that the system is easy to use and navigate

Summary of Survey Results (Weighted Mean)

Statement	**Mean**	**Interpretation**
1.The system is easy to use and navigate	3.83	Agree
2.The interface is clear and organized	3.46	Agree
3.Easy to find system features	3.8	Agree
4.System accurately records data	3.81	Agree
5.Reports are useful and complete	3.82	Agree
6.Features meet poultry management needs	3.77	Agree
7.System performs tasks quickly	3.76	Agree

8.Reduces manual work and saves time	3.82	Agree
9.System ensures data security	3.82	Agree
10.Over All Satisfaction	3.81	Agree

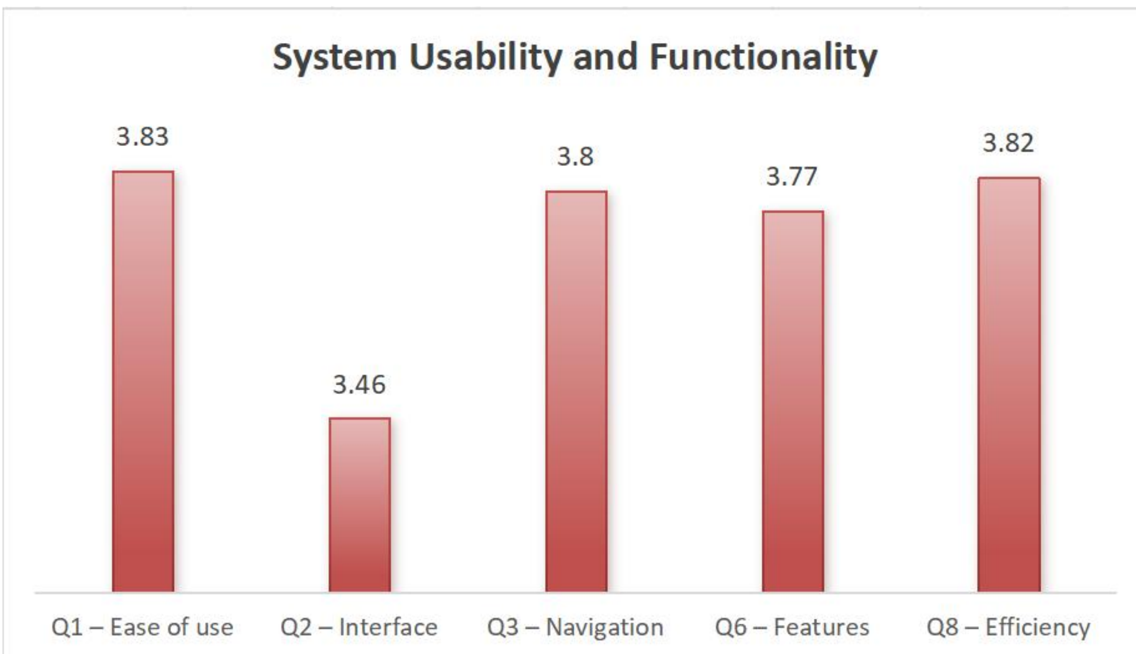
We use the weighted mean in our survey because we selected the Likert scale for the survey, where each response is assigned a numerical value. This method enables us to determine an overall average by considering both the response values and their frequencies. It provides a clearer understanding of the respondents' level of agreement.

Graphical Representation of Data

Figure 1.1: System Performance, Security, and Satisfaction

This figure shows the assessment of the system based on accuracy, report generation, performance, security, and overall user satisfaction. These indicators show how reliable and effective the system is for managing poultry.

Figure 1.2: System Usability and Functionality



This figure shows how usable the system is. It covers ease of use, interface design, navigation, system features, and efficiency. The results reveal how users interact with the system and what they gain from it.

System Performance and Testing

System tested was conducted by both functional testing and user testing to assess the overall performance of **Integrated Livestock Ordering System (ILOS)**

Functional Testing:

All system components, including data management, ordering process, and conduct the inventory and were found properly function

User testing:

This involved a select group of respondents who engaged directly with the system. Their feedback was collected to assess usability, efficiency, and overall system performance.

Performance Results:

- The system showed a quick response time
- Accurate data recording
- Reduced manual errors
- High usability rating

These findings are backed by the weighted mean results, where all indicators received an "Agree" rating.

User Feedback, Testing Results, and Accuracy Rates

Positive Feedback:

- System is easy to use
- Saves time and reduces workload
- Improves accuracy of records

Constructive Feedback:

- Needs minor interface improvements
- Requires user training for new users

Overall, most respondents rated the system as **satisfactory**

Summary of Significant Results

1. System improved efficiency of operations
2. Accuracy of data recording increased

3. User satisfaction levels were high
4. Manual errors were significantly reduced
5. System supports better decision-making

Analysis and Interpretation

The analysis shows that the ILOS system effectively improves poultry management processes.

- **Efficiency:** Faster task completion
- **Accuracy:** Reduced human errors
- **Usability:** Easy to learn and operate

The system successfully addresses common issues in traditional poultry management.

Explanation of Trends, Patterns, and Comparative Evaluation

Trends and Patterns:

- Most responses fall under "Agree"
- Few "Neutral" responses are present
- Users prefer automated processes
- Satisfaction increases with system usage

Comparative Evaluation:

Aspect	**Traditional**	**ILOS**
Data Recording	Manual	Automated
Speed	Slow	Fast
Accuracy	Moderate	High
Efficiency	Low	High

Identification of Strengths and Limitations

Strengths:

- Easy to use
- Reduces workload
- Improves accuracy
- Secure system

Limitations:

- Requires basic technical knowledge
- Needs continuous updates

Impact of ILOS on the Information Technology Field

The creation of the castillano backyard : integrated livestock ordering system significantly contributed to the domain of Information Technology by bringing up-to-date web technologies together with agricultural business processes. This system indicates how IT can be applied to automate livestock management and establish an effective online platform for selling, filling the gap between e-commerce and traditional poultry farming. By utilizing database management, user authentication, and online transaction modules, the system illustrates how technology can simplify inventory monitoring, sales tracking, and customer interaction processes.

In the IT world, the system aids innovation and digitalization in the agriculture and animal husbandry industry. The system is also a demonstration of how information systems can be built to assist poultry farmers and poultry owners through automated activities and data-based decisions. The incorporation of online features like live product listing, automated sales calculation, and online payment gateways demonstrates the utilitarian use of IT in developing intelligent and efficient business solutions.

Additionally, the system facilitates technological innovation and professional development for IT practitioners and students through the encouragement of innovation that fuses web development, database design, and cybersecurity concepts. It illustrates the relevance of IT in enhancing sustainable agriculture and developing platforms that enhance accessibility for sellers and buyers.

In general, the system supports the continuous growth of the Information Technology field by showing how technology can change traditional industries, enhance productivity, and meet the rising need for online business solutions in managing agriculture and livestock.

Project Structure

```
poultry/
├── admin/
│   ├── uploads/
│   │   └── products/
│   │
│   ├── admin_dashboard.php
│   ├── admin_login.php
│   ├── admin_profile.php
│   ├── orders.php
│   ├── products.php
│   ├── process.php
│   ├── conn.php
```

```
│   ├── header.php
│   └── test_sms.php
│
├── customer/
│   ├── uploads/
│   │
│   ├── cart.php
│   ├── checkout.php
│   ├── customer_profile.php
│   ├── shop.php
│   ├── process.php
│   ├── conn.php
│   └── header.php
│
├── css/
│   ├── bootstrap.min.css
│   └── style.css
│
├── js/
│   └── main.js
│
├── .env
├── LICENSE.txt
├── composer.json
├── composer.lock
│
├── conn.php
├── process.php
├── header.php
```

```
├── footer.php
|
├── index.php
├── login.php
├── logout.php
├── register.php
└──contact.php
```

Chapter 5

Summary

The Castillano's Backyard: Integrated Livestock Ordering System (ILOS) was developed to streamline and modernize the process of selling livestock and related products such as pigs, chickens, eggs, and artificial insemination services. The system provides a simple yet efficient web-based platform that allows customers to conveniently browse products, place orders, and track their transactions online.

The project employs a two-tier architecture, consisting of a presentation layer (web interface) and a data layer (MySQL database). The web interface, built with HTML, CSS, JavaScript, and Bootstrap, ensures a user-friendly experience for both customers and the admin. Meanwhile, the database securely stores and manages product, order, inventory, and sales information.

Through the implementation of ILOS, manual processes such as product listing, order tracking, and sales computation are automated. This results in improved accuracy, faster transaction times, and better inventory monitoring. The system also provides real-time sales updates and product availability, which helps the owner make more informed business decisions.

In conclusion, the development of the Integrated Livestock Ordering System demonstrates how digital solutions can support small-scale backyard businesses in adopting e-commerce technology. It bridges the gap between traditional selling methods and modern online transactions, improving operational efficiency and overall customer satisfaction.

Conclusion and Recommendation

Conclusion

The creation of the Castillano's Backyard: Integrated Online Livestock Ordering System has been an efficient means of enhancing the management and efficiency of a small-scale livestock business. With this system, the owner of the Castillano’s Backyard can readily track inventory, handle customer orders, and track sales and profit, all within one integrated system.

The system offers an easy platform for customers to view available livestock products and make orders online, saving them the effort of manual transactions and the risk of inaccurate records. Largely because it's a niche and rare business idea — selling live animals online — the system proves that even conventional businesses can gain significantly from digital solutions.

This project shows the way technology can assist local and small-scale businesses in making their operations more modern, accessing more customers, and running their business better.

Recommendation

The researchers recommend the integration of an **Application Programming Interface (API)** into the system to enhance connectivity and accessibility. By implementing an API, there will be easy communication between the online livestock ordering portal and other applications or services, making it easy for customers to access the e-commerce capabilities of Castillano's Backyard more efficiently.

Also, inclusion of Machine Learning (ML) functions is strongly recommended. This would assist the admin in automatically processing and computing monthly sales, profits, and finance trends. With the utilization of ML-based analytics, the system can offer insights based on data that help in informed business decisions and enhance overall farm management.

Additional future improvements like integrating online payments, mobile app creation, and real-time analysis of data are also suggested to further enhance system performance, accessibility, and user satisfaction.

References (APA Format)

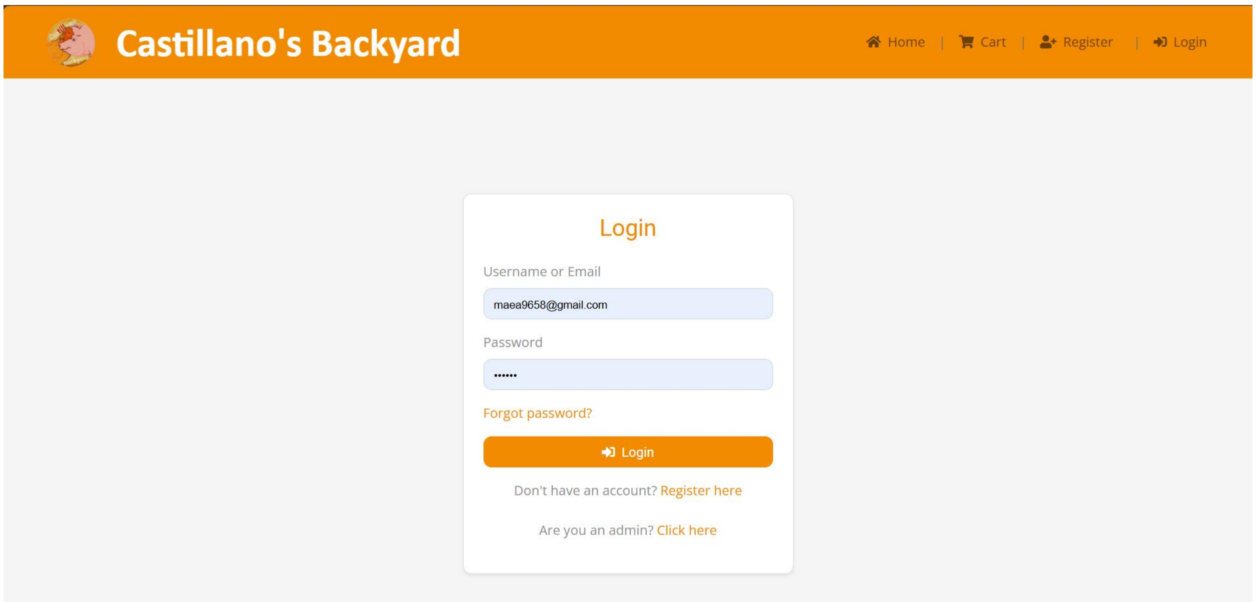
Haag, S., & Cummings, M. (2009). *Management information systems for the information age* (7th ed.). McGraw-Hill Higher Education. ISBN 978-0-07-128796-8

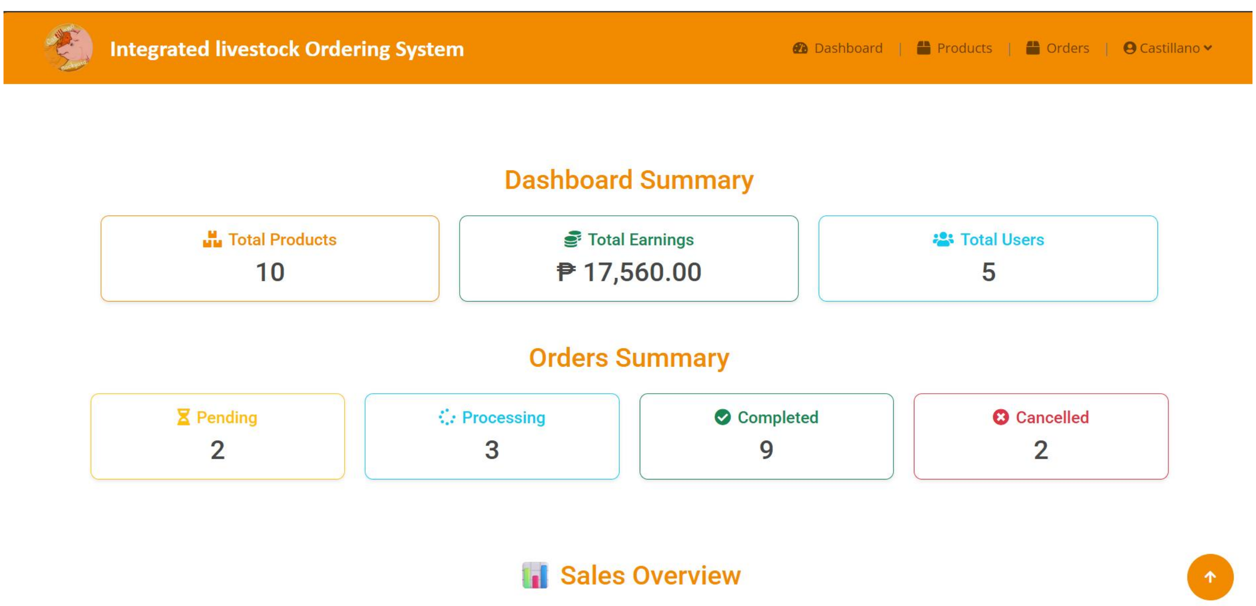
Rahman, M. A., & Casonovas, L. (2021). Strategies to predict e-commerce inventory and ordering planning. In M. Khosrow-Pour (Ed.), Handbook of Research on Intelligent Techniques and Modeling Applications in Marketing Analytics. ISBN 1799889580

Appendices:

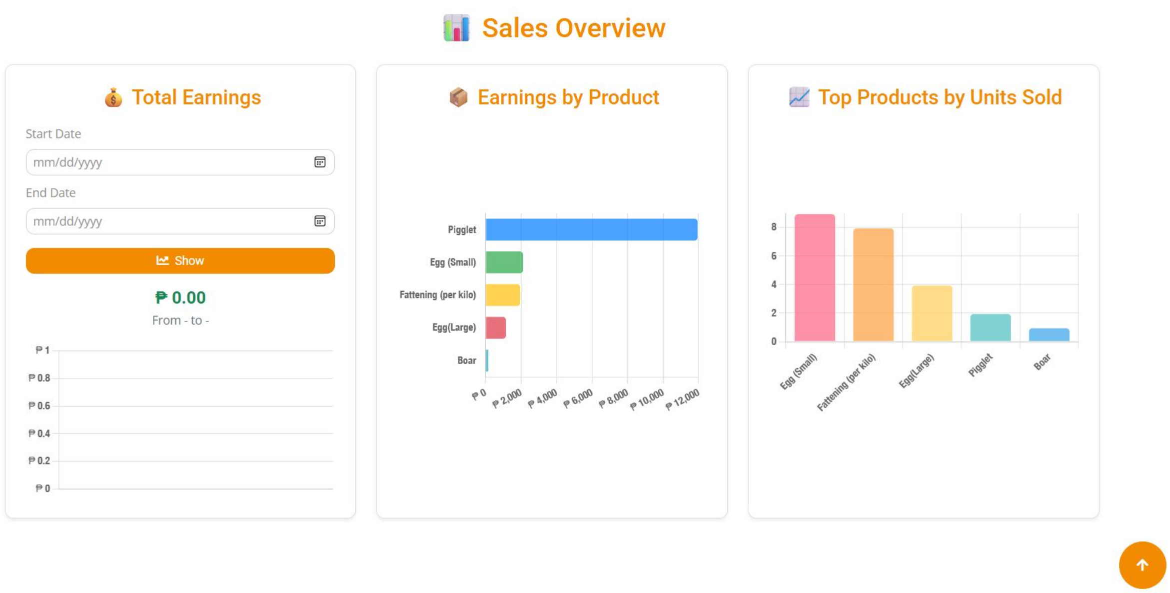
Appendix A: System Interface Screenshots

- Login Page



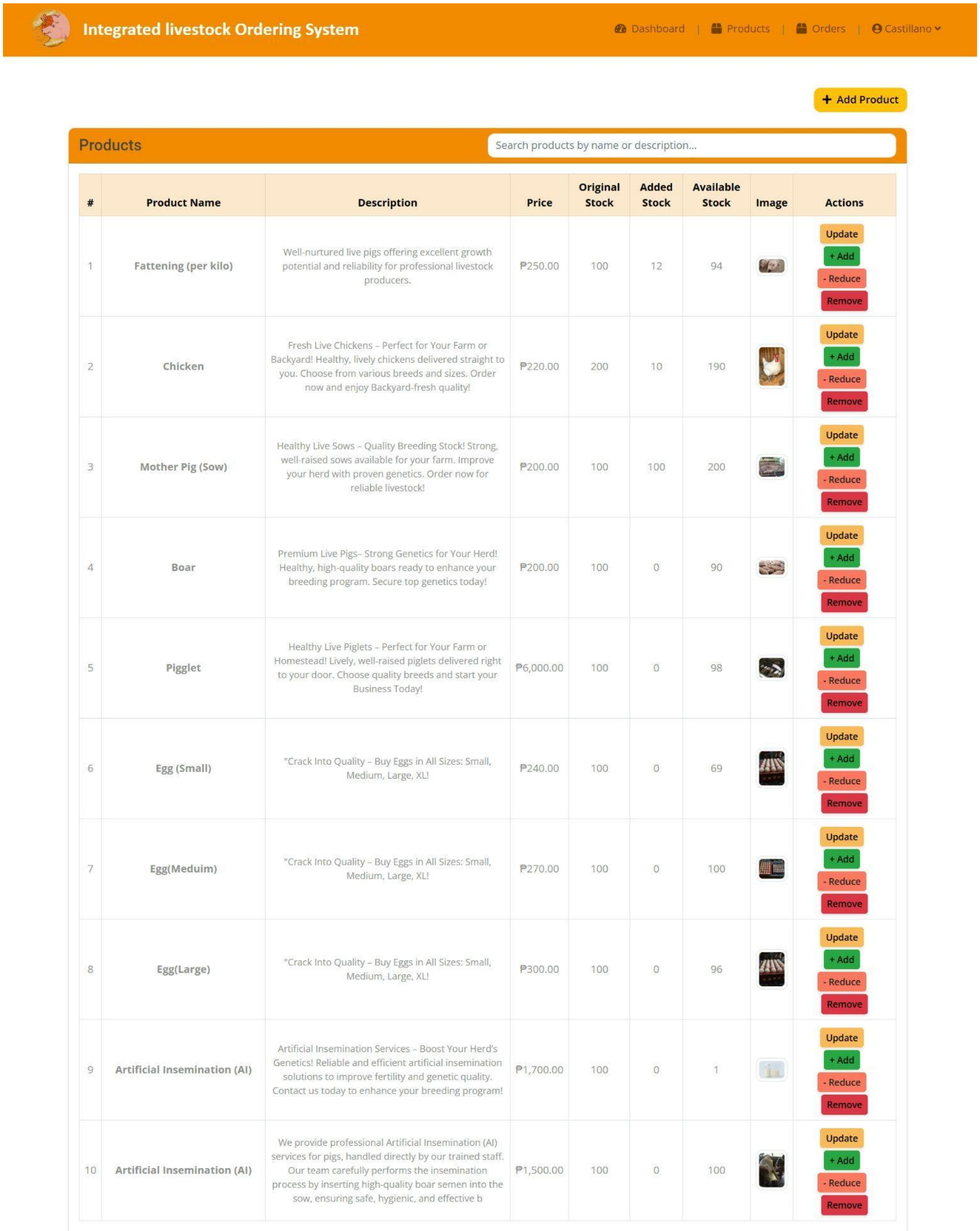


- Dashboard


Sales Overview
Total Earnings
Start Date
mm/dd/yyyy
End Date
mm/dd/yyyy
Show
₱ 0.00
From - to -
Earnings by Product
Pigglet
Egg (Small)
Fattening (per kilo)
Egg(Large)
Boar
Top Products by Units Sold
Egg (Small)
Fattening (per kilo)
Egg(Large)
Pigglet
Boar

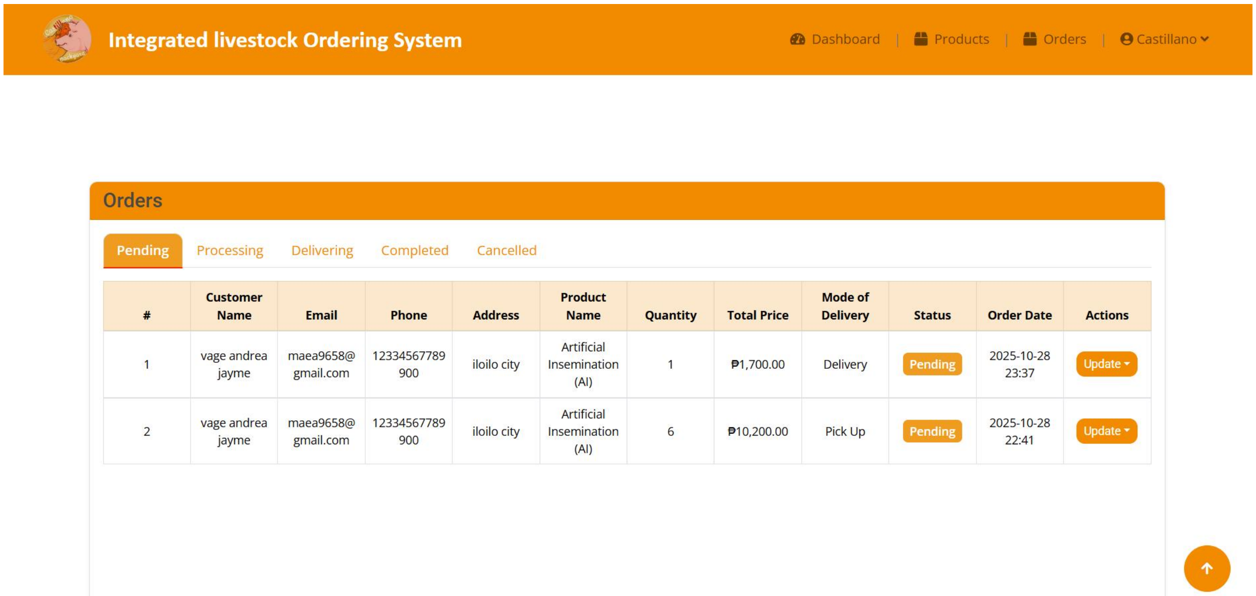
-

- Products



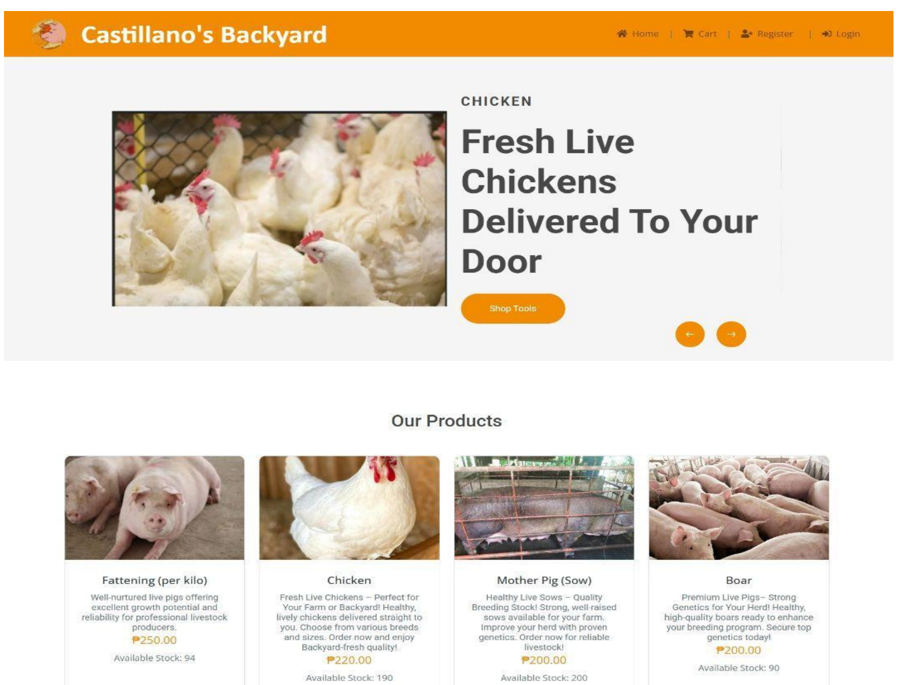
#	Product Name	Description	Price	Original Stock	Added Stock	Available Stock	Image	Actions
1	Fattening (per kilo)	Well-nurtured live pigs offering excellent growth potential and reliability for professional livestock producers.	₱250.00	100	12	94		Update + Add - Reduce Remove
2	Chicken	Fresh Live Chickens – Perfect for Your Farm or Backyard! Healthy, lively chickens delivered straight to you. Choose from various breeds and sizes. Order now and enjoy Backyard-fresh quality!	₱220.00	200	10	190		Update + Add - Reduce Remove
3	Mother Pig (Sow)	Healthy Live Sows – Quality Breeding Stock! Strong, well-raised sows available for your farm. Improve your herd with proven genetics. Order now for reliable livestock!	₱200.00	100	100	200		Update + Add - Reduce Remove
4	Boar	Premium Live Pigs– Strong Genetics for Your Herd! Healthy, high-quality boars ready to enhance your breeding program. Secure top genetics today!	₱200.00	100	0	90		Update + Add - Reduce Remove
5	Pigglet	Healthy Live Piglets – Perfect for Your Farm or Homestead! Lively, well-raised piglets delivered right to your door. Choose quality breeds and start your Business Today!	₱6,000.00	100	0	98		Update + Add - Reduce Remove
6	Egg (Small)	"Crack Into Quality – Buy Eggs in All Sizes: Small, Medium, Large, XL!	₱240.00	100	0	69		Update + Add - Reduce Remove
7	Egg(Meduim)	"Crack Into Quality – Buy Eggs in All Sizes: Small, Medium, Large, XL!	₱270.00	100	0	100		Update + Add - Reduce Remove
8	Egg(Large)	"Crack Into Quality – Buy Eggs in All Sizes: Small, Medium, Large, XL!	₱300.00	100	0	96		Update + Add - Reduce Remove
9	Artificial Insemination (AI)	Artificial Insemination Services – Boost Your Herd's Genetics! Reliable and efficient artificial insemination solutions to improve fertility and genetic quality. Contact us today to enhance your breeding program!	₱1,700.00	100	0	1		Update + Add - Reduce Remove
10	Artificial Insemination (AI)	We provide professional Artificial Insemination (AI) services for pigs, handled directly by our trained staff. Our team carefully performs the insemination process by inserting high-quality boar semen into the sow, ensuring safe, hygienic, and effective b	₱1,500.00	100	0	100		Update + Add - Reduce Remove

- Orders

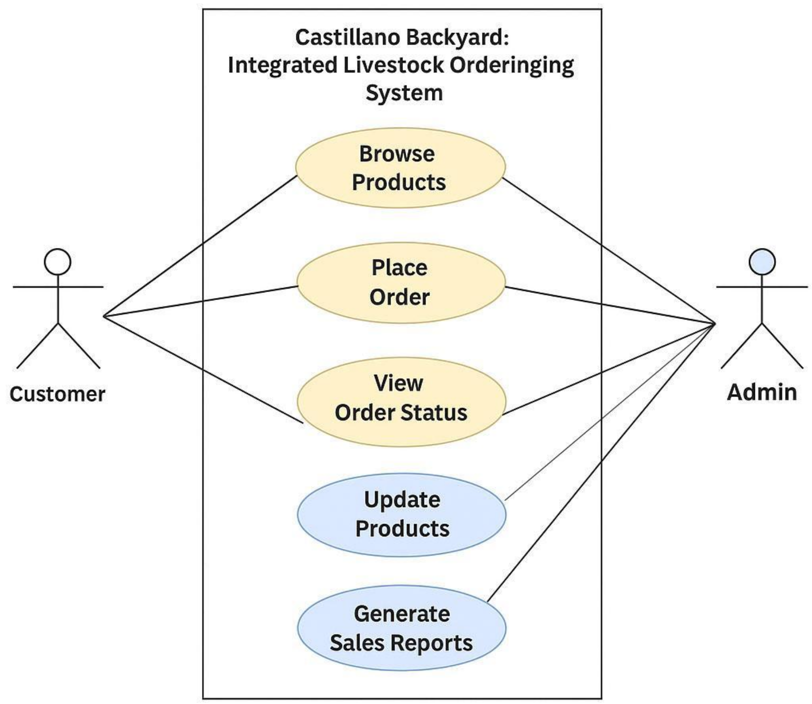


#	Customer Name	Email	Phone	Address	Product Name	Quantity	Total Price	Mode of Delivery	Status	Order Date	Actions
1	vage andrea jayme	maea9658@gmail.com	12334567789900	iloilo city	Artificial Insemination (AI)	1	₱1,700.00	Delivery	Pending	2025-10-28 23:37	Update
2	vage andrea jayme	maea9658@gmail.com	12334567789900	iloilo city	Artificial Insemination (AI)	6	₱10,200.00	Pick Up	Pending	2025-10-28 22:41	Update

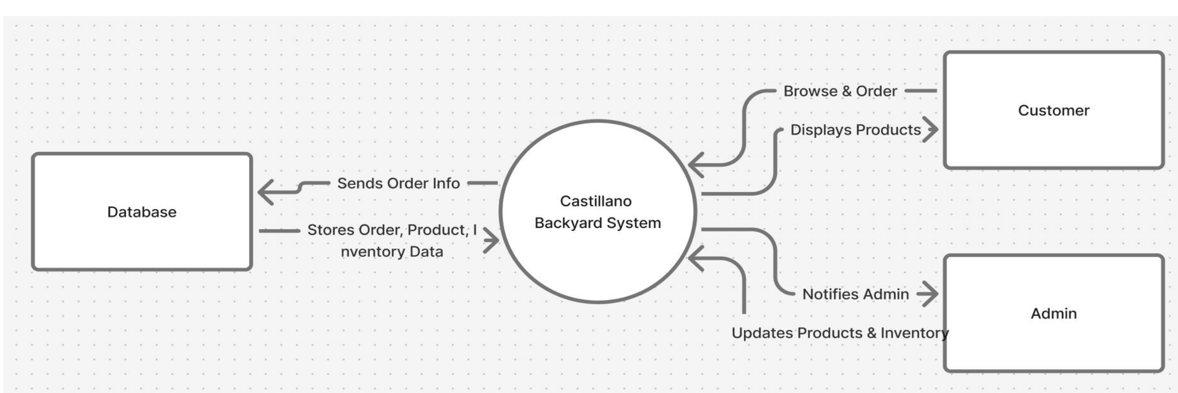
Customer side

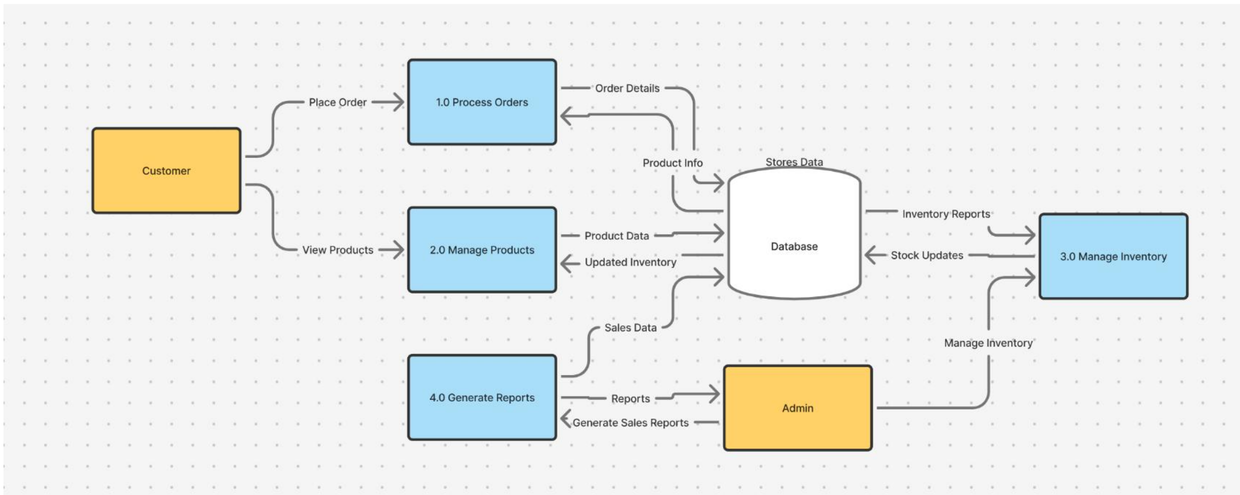


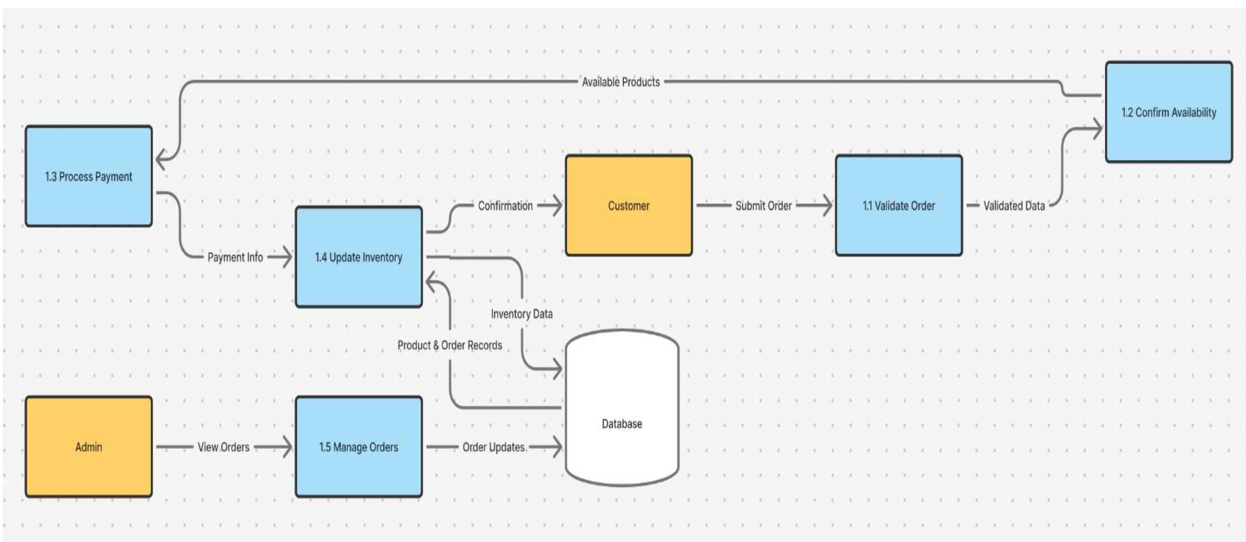
Appendix B: Use Case Diagram



Appendix C: Data Flow Diagram







Appendix D: User Survey Form and Summary

Demographic Information

Name: ____________________

Age:

☐ 18-25 ☐ 26-35 ☐ 36-45 ☐ 46 and above

Gender:

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☐ Male ☐ Female ☐ Prefer not to say ☐ Other

Frequency of Using the System:

☐ Daily ☐ Weekly ☐ Occasionally ☐ Rarely

Please rate your level of agreement with the following statements based on your experience using the system.

Scale:

5 – Strongly Agree 4 – Agree 3 – Neutral 2 – Disagree 1 – Strongly Disagree

___1. The system is easy to use and navigate.

___2. The interface is clear and visually organized.

___3. I can easily find the features or functions I need.

___4. The system accurately records poultry data such as feeds, stocks, and production.

___5. The reports generated by the system are useful and complete.

___6. The system's features meet the needs of poultry management.

___7. The system performs tasks quickly without delay or error.

___8. Using the system reduces manual work and saves time.

___9. The system keeps my information and account secure from unauthorized access.

___10. Overall, I am satisfied with the ILOS and would recommend it to others.

What feature of the system do you find most useful? (Optional)

____

What problems or challenges have you encountered while using the system?(Optional)

____

What suggestions do you have for improving the system?(Optional)

____

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+63 909 370 2750
Rizal Pala-Pala Zone 1
Iloilo City, 5000

Martin Castillano Jr.
Castillano's Backyard ILOS
Tapaz, Capiz
April, 2026

Dear Mr. Castillano,

Subject: Proposal for Memorandum of Agreement to Use the "Castillano's Backyard Integrated Livestock Ordering System (ILOS)"

I am pleased to present the proposed Castillano's Backyard: Integrated Livestock Ordering System (ILOS), a digital platform designed to automate and enhance the livestock management and online selling process. The system aims to streamline operations through automated recording, real-time tracking, and efficient online transactions, ensuring better data accuracy and improved productivity for the business.

To formalize the implementation and use of this system, we propose the creation of a Memorandum of Agreement (MOA) outlining the responsibilities, terms, and conditions of both parties. This agreement will establish a clear understanding regarding project scope, deployment timeline, data security, system support, and maintenance.

The ILOS system is expected to deliver the following benefits:

- Efficient and accurate livestock record management – reducing human errors through automation.
- Online selling platform – enabling customers to view and purchase livestock conveniently.

- Real-time monitoring and updates – allowing the owner to track sales, inventory, and animal status.
- User-friendly interface – simple and accessible for both administrators and buyers.
- Secure data handling – ensuring confidentiality and protection from unauthorized access.
- Automated notifications and reports – for tracking livestock sales, health status, and transactions.

Our team is committed to providing full technical support, including training and maintenance, to ensure smooth implementation and optimal system performance. The goal of this project is to strengthen Castillano's Backyard's digital transformation, making it a model of innovation within the agricultural and livestock industry.

We look forward to discussing this proposal further and working together for the successful adoption of ILOS. Kindly inform us of a convenient time to meet or if additional details are required.

Thank you for your consideration.

Sincerely,

____

Mae Christil Joy Sumbing
Programmer
ILOS

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MEMORANDUM OF AGREEMENT (MOA)

Between
Castillano’s Backyard ILOS
And
Martin Castillano Jr., Mae Christil Joy Sumbing, and the Project Development Team

Phoebe Mae Almario
Mary Mae Babac
Kenth Axell Castillano
Mae Christil Joy Sumbing
Date: April, 2026

Purpose

This Memorandum of Agreement (MOA) is made and entered into by and between Castillano’s Backyard ILOS, located at *Tapaz, Capiz*, and the Project Development Team led by Maee Christil Joy Sumbing from *Rizal Pala-Pala Zone 1, Iloilo City*. The purpose of this agreement is to formalize the collaboration for the development and implementation of the project entitled “Castillano’s Backyard Livestock Online System (ILOS)”.

The system aims to automate livestock management, enable online sales transactions, and enhance data accuracy and efficiency in managing poultry and livestock operations.

Scope of Agreement

Castillano’s Backyard ILOS agrees to provide the following:

- Access to relevant information and existing livestock management processes.
- Feedback and guidance during the system development and testing phases.

- Permission to utilize data and records necessary for the project's implementation.

The Project Development Team, led by Maee Christil Joy Sumbing, agrees to:

- Conduct research, system design, development, and testing.
- Ensure the system meets the operational needs of Castillano's Backyard.
- Maintain confidentiality and security of all data shared by the client.
- Deliver the completed system according to the agreed timeline and requirements.

Client Approval

By signing this agreement, Castillano's Backyard ILOS acknowledges approval and support for the Castillano's Backyard Livestock Online System (ILOS) project. Both parties agree to cooperate in achieving the successful development, testing, and deployment of the system. The client also commits to providing timely feedback, access, and relevant resources for the project's completion.

Signatures:

For the Castillano's Backyard:

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PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

Martin Castillano Jr.
Owner, Castillano's Backyard
Tapaz, Capiz
Date: ___________________

For the Project Development Team:

Mae Christil Joy Sumbing
Programmer
Rizal Pala-Pala Zone 1, Iloilo City
Date: ___________________

Phoebe Mae Almario
UI/UX
Date: ___________________

Mary Mae Babac
Documentor
Date: ___________________

Kenth Axell Castillano
Documentor
Date: ___________________